

BROKILD · WINDOWS VST3 · SNARE DRUM SYNTHESISER

X SNARE TACTICS

A head, a set of wires and a stick. Every snare there is, from those three.
No samples.

MANUAL · BUILD 260926.4

The interface features a timeline at the top with markers from 0 ms to 350 ms. Below the timeline are three main sections: HEAD, WIRES, and HIT. The HEAD section includes controls for TUNE (190 Hz), DROP (3.0 st), BEND (12 ms), DECAY (280 ms), WAVE (0 %), SKIN (90 %), RING (35 %), and BODY (80 %). The WIRES section includes controls for WIRES (60 %), SIZZLE (320 ms), TENSION (55 %), and AIR (75 %). The HIT section includes controls for STRIKE (30 %), STICK (40 %), CLAP (0 %), and SPREAD (45 %). Below these are four additional sections: ERA (GRIT, ROOM, GATE), ECHO (ECHO, TIME, 1/8D, 120 BPM), TRANSIENT (ATTACK, SUSTAIN), and DRIVE (ENGINE, DRIVE, COLOUR, IDLE BURN, HYPERDRIVE, RAZOR WING, SUPERNOVA). The COMP section includes controls for COMP (35 %) and SPEED (60 %). The OUT section includes controls for LEVEL (+1.6 dB) and VELO (50 %).

0 ms 50 ms 100 ms 150 ms 200 ms 250 ms 300 ms 350 ms

head: snare hoop: rim shot

RIM X-STICK CLAP

HEAD

TUNE DROP BEND DECAY WAVE SKIN RING BODY

190 Hz 3.0 st 12 ms 280 ms 0 % 90 % 35 % 80 %

WIRES

WIRES SIZZLE TENSION AIR

60 % 320 ms 55 % 75 %

HIT

STRIKE STICK CLAP SPREAD

30 % 40 % 0 % 45 %

ERA

GRIT ROOM GATE

0 % 20 % OFF

ECHO

ECHO TIME

1/8D 120 BPM

0 %

TRANSIENT

ATTACK SUSTAIN

+20 % 0 %

DRIVE

ENGINE DRIVE COLOUR

IDLE BURN HYPERDRIVE RAZOR WING SUPERNOVA

0 % 100 %

COMP

COMP SPEED

35 % 60 %

OUT

LEVEL VELO

+1.6 dB 50 %

KEYS GM KIT

WHAT IS IN HERE

03	QUICK START	Install it, hit it, pick a preset.
04	THE IDEA	Three sources, coupled.
05	HEAD	The drum itself: eight knobs.
06	WIRES	The snare in the snare drum.
07	HIT	Where, with what, and the clap.
08	ERA & ECHO	The machine, the room, the gate, the dub.
09	TRANSIENT & DRIVE	Front and back; four engines.
10	COMP & OUT	Crack or pump; level, touch and keys.
11	THE DISPLAY & THE PAD	Head, wires, pitch, and four ways to hit.
12	RECIPES	From one style to another.
13	PRESETS	Forty-four, in seven banks.
15	SPECIFICATIONS	And what the bench measured.

IN SHORT

Snare Tactics synthesises snare drums: the studio snare, the drum machines that made the records, and what techno, dub, industrial and digital hardcore did with them next.

It does not carry a simulation per machine. It models what every snare is — a head, wires and a stick, coupled — with thirty-one controls, and lets you walk from one drum to the next.

THE FIRST TEN SECONDS

Open it and click the **HIT** pad — then the chrome hoop around it for a rim shot. Press > beside the preset name until something makes you grin.

INSTALL, HIT, PLAY



The panel. The display and the pad across the top. Below, the controls in the order the sound meets them: Head, Wires, Hit, then Era, Echo, Transient, Drive, Comp and Out.

1 • INSTALL

Copy **Snare Tactics.vst3** into C:\Program Files\Common Files\VST3\ and rescan. It files under Instrument / Drum. The standalone, **Snare Tactics.exe**, needs no installing.

2 • FIRST HIT

Click the **HIT** pad — the centre is the hardest hit — or its hoop for a rim shot. Keys: **space** snare, **R** rim shot, **X** cross-stick, **C** clap.

3 • PICK A PRESET

< and > step through the banks; the name opens the list. Presets are level-matched, so you compare drums, not volumes.

4 • FROM MIDI

With **KEYS** on GM KIT (the default), note 38 is the snare, 40 a rim shot, 37 a cross-stick and 39 the clap — the General MIDI drum map, so a drum rack just works.

5 • KEEP IT

SAVE and **LOAD** use Documents\Brokild patches\Snare Tactics.

THREE SOURCES, COUPLED

A snare is a head, a set of wires and a stick — and they are not independent. The wires are driven by the head. That coupling is most of what makes a real snare sound real.

WHAT	808 / 909	STUDIO SNARE	SNARE TACTICS
The head	two oscillators, 1 : 1.8	a membrane’s modes	SKIN walks between them
Where it is hit	—	centre, edge, rim	STRIKE picks the modes
The wires	filtered noise	driven by the head	lag, buzz and ring along
Wire tension	—	loose to tight	TENSION
The stick	—	click, rim, cross-stick	STICK, STRIKE, articulations
Hands	the clap	—	CLAP and SPREAD

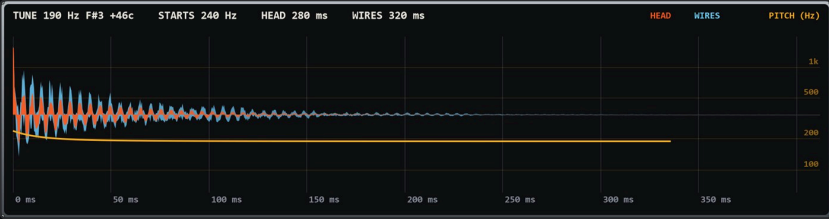
A drum machine’s noise ignores the head. Real wires start a fraction of a millisecond after the stick, slap against the head once a cycle when they are loose, and keep sizzling for as long as the head rings. Snare Tactics does all three — which is why a loose snare here buzzes **at the drum’s own pitch**.



808 Snare. Two tones and a little noise.



909 Snare. Triangles, the drop, bright noise.



Studio Snare. A membrane, wires, a stick.

HEAD

TUNE 100–420 Hz

The head's settled pitch. The display names the note, so the snare can sit in the key of the song.

DROP 0–24 semitones

How far above TUNE the hit starts. A few semitones is a real head being hit hard; a lot is a zap.

BEND 2–200 ms

How fast the pitch falls back. Short is a snap; long, with a big DROP, is the Simmons sweep.

DECAY 40 ms–2 s

Time for the head to fall 60 dB. The wires have their own.

WAVE sine → square

The wave of the electronic pair: sine (808), triangle (909, at 50 %), rounded square. The membrane is always sines.

SKIN 0–100 %

At 0, the drum-machine body: two oscillators near 1 : 1.83. At 100, a real membrane — ten modes at the Bessel ratios, and a pitch that starts sharp when hit hard and then settles.

RING 0–100 %

How undamped the head is. Low is gel and tape: the overtones die at once. High leaves the famous snare ring wide open.

BODY 0–100 %

How much of the head you hear. All the way down for a clap or the wires on their own.

WIRES

The steel coils under the bottom head — the snare in the snare drum.

WIRES

Their level. The 808 called this SNAPPY.

SIZZLE 30 ms–2 s

Time for the wires to fall 60 dB. It also sets how long a clap hangs.

TENSION loose → tight

The one that makes it a real snare. **Loose** wires are dark, start late, slap against the head once a cycle — a buzz at the drum's own pitch — and keep ringing along with the head. **Tight** wires are bright, short and crisp.

AIR

The top end of the wires and of the clap.

WHAT TENSION DOES, MEASURED

From loose to tight the wires' brightness (spectral centroid) rises from **3.7 kHz** to **7.8 kHz**. Loose, their noise is chopped at the head's pitch **45 dB** more than tight; they arrive **3.2 ms** later; and with a long-ringing head they are still sizzling when tight wires have long gone.

GHOST NOTES

Play softly and the balance tips toward the wires, the way it does on a real drum: a ghost note is mostly snare, not head. With VELO up that is **7 dB** of difference in the mix between a ghost and a full hit.



The wires in steel, the head in orange. Big Eighties: loose-ish wires into a gated room.

STRIKE centre → rim

Where the stick lands. A centre hit can only excite the head’s round-symmetric modes; out toward the edge the others join in and the ring grows; past 70 % it becomes a rim shot, with the shell and hoop ringing too.

STICK

The stick’s own click.

CLAP

An 808-style hand clap layered on the snare: noise through a band-pass, several bursts, then a tail.

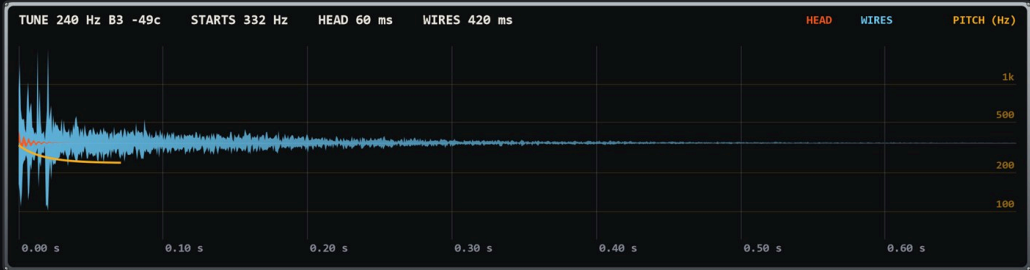
SPREAD

The clap’s burst spacing. At 0, one tight burst; up from there, four hands a little out of time with each other.

FOUR WAYS TO HIT IT

ARTICULATION	GM NOTE	PAD / KEY	WHAT CHANGES
Snare	38	head · space	as the controls say
Rim shot	40	hoop · R	edge and rim, louder, the shell rings
Cross-stick	37	X-STICK · X	head choked, wires still, shell only
Clap	39	CLAP · C	the clap alone

All four come from the same controls, so a patch is a whole snare, not one sound. A rim shot measured +15 dB of the shell’s 520 Hz mode over a centre hit; a cross-stick carries 22 dB less wire.



Warehouse Clap. Four bursts at SPREAD 55 %, then the tail.

ERA & ECHO

ERA • THE MACHINE AND THE ROOM

GRIT

Lower sample rate and bit depth. Around **30 %** is the 12-bit SP-1200 and Akai; around **55 %** the 8-bit Linn and DMX; beyond that, an Atari.

ROOM

A room behind the drum; more is bigger as well as louder. It sits before the drive and the compressor, so it can be crushed.

GATE **OFF**, 20-600 ms

The 80s gate. The room — and the drum — is cut this long after each hit. It is keyed by the hit itself, not by a threshold, so it opens on the stick and closes exactly on time.

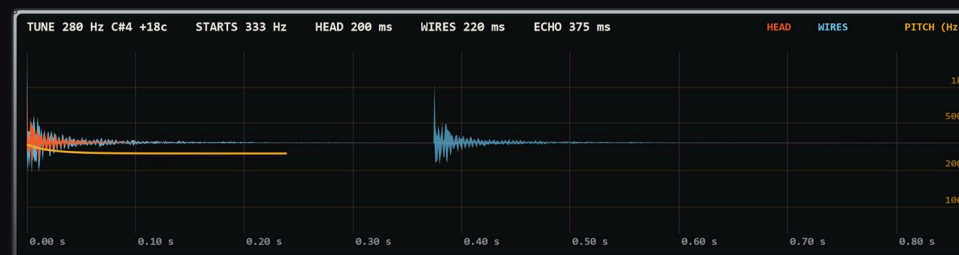
ECHO • THE DUB

ECHO

A tape echo, ping-pong: the first repeat on the left, the next on the right. One knob raises the send and the feedback together. Each pass comes back darker and thinner, and saturates, so it can be pushed to the top without running away.

TIME **1/16** ... **1/2**

Locked to the host tempo, which is shown beneath it; 120 BPM when the standalone has no host. Dotted eighth is the default for a reason.



Tubby Throw. A rim shot and its first repeat, 375 ms later — a dotted eighth at 120.

TRANSIENT & DRIVE

TRANSIENT

ATTACK bipolar

More or less of the first milliseconds: the crack.

SUSTAIN bipolar

More or less of the tail: the wires and the room.

A transient shaper usually has to guess from the audio. Snare Tactics is the synthesiser, so it reads its own envelopes — head, wires, stick and clap — and the gain it makes cannot ripple.

DRIVE • FOUR ENGINES FROM BATTLESTAR OVERDRIVE

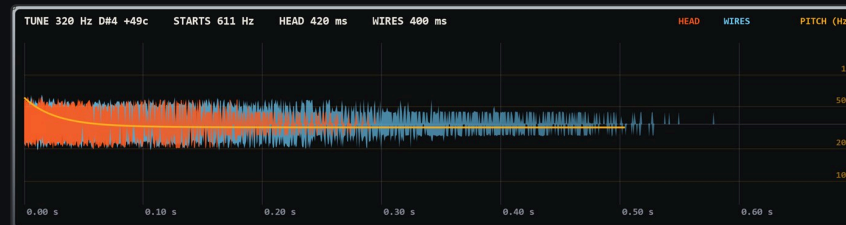
ENGINE	WHAT IT DOES
IDLE BURN	A warm, asymmetric tube. Thickens without shouting.
HYPERDRIVE	Harmonic injection: an EBM snap, a snare that cuts on small speakers.
RAZOR WING	Tanh into a hard clip. Gabber, breakcore, hard techno.
SUPERNOVA	Fold, crush and chaos. Digital hardcore, industrial.

DRIVE

How hard the engine is pushed. Level-matched by measurement: more drive is more character, not more volume.

COLOUR

The drive's low-pass. Fully open is no filter at all.



Metal Shop. A ringing rim struck hard, through SUPERNOVA.

COMP & OUT

COMP

COMP

Threshold and ratio on one knob, with its own make-up, so turning it up makes the snare denser, not quieter.

SPEED

Slow lets the crack through before the compressor grabs the body. Fast flattens it and brings the wires and room up to meet the hit. It looks 1.5 ms ahead, so even fast it catches the front.

ROLLS

Four hits sound at once and they overlap the way a real drum does. A fifth fades the oldest over 3 ms rather than cutting it, so a 32nd-note roll at 170 BPM stays smooth.

OUT

LEVEL

Output. A soft ceiling holds everything under 0 dBFS.

VELO

How much harder hits are louder, brighter, and further from a ghost note. At 0, every velocity is the same hit, exactly.

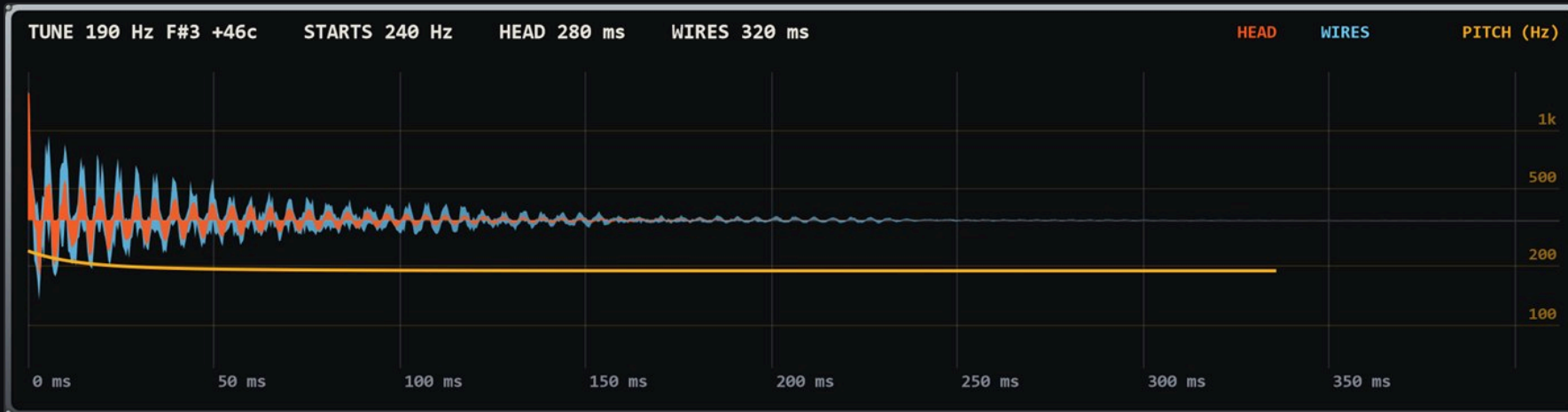
KEYS

FIXED: every note is the snare. **GM KIT:** 38 snare, 40 rim shot, 37 cross-stick, 39 clap, everything else the snare. **CHROMATIC:** the note tunes the head, with D1 (38) at TUNE — a tuned snare line, or a pitched rim melody.

STEREO

The drum is mono, both sides the same sample for sample. The only thing that spreads it is the echo, ping-ponging.

THE DISPLAY & THE PAD



Studio Snare. The head in orange over the whole drum in steel, the head's pitch in amber; the readouts along the top.

HEAD & WIRES

The hit is rendered twice every time a control moves: the head on its own in orange, and the whole drum in steel behind it. Whatever is steel and not orange is the wires, the stick, the clap and the room.

THE READOUT

TUNE and its note, where the pitch starts, how long the head and the wires take to fall 60 dB, and the echo time at the current tempo.

PAD & METERS

The head is the snare, the centre hardest. The chrome hoop is a rim shot. RIM, X-STICK and CLAP below it. OUT and GR meters beside.

RECIPES

From one style to another, a few knobs at a time.

808 → 909

1. Start on **808 Snare**.
2. WAVE to 50 %: triangles.
3. DROP up to about 7, BEND 10 ms.
4. TENSION and AIR up: brighter, tighter noise.

MACHINE → REAL DRUM

1. SKIN up to 90 %.
2. STRIKE to 30 %, RING to taste.
3. TENSION down a little, so the wires buzz.
4. STICK up, a little ROOM.

THE BIG 80s SNARE

1. A studio snare, STRIKE 60 %.
2. ROOM up to 75 %.
3. COMP 55 %, SPEED fast: the room comes up.
4. GATE to 240 ms. Done.

DUB THROW

1. Tune up high, STRIKE to 100 %: a rim shot.
2. ECHO up, TIME 1/8D.
3. COLOUR down a touch.
4. Ride ECHO from the host: throw it on the last hit of the bar.

DIGITAL Hardcore

1. A break snare: SKIN 80 %, STRIKE 50 %.
2. GRIT to 60 %.
3. SUPERNOVA, DRIVE 80 %.
4. COMP 60 %, SPEED fast. Roll it at 180.

SNARE + CLAP STACK

1. Any electronic snare.
2. CLAP up to 70 %, SPREAD 50 %.
3. SIZZLE sets how long the clap hangs.
4. A little ROOM behind both.

PRESETS

ACOUSTIC

Studio Snare

A 14-inch wooden snare, mic'd close, a little room.

Piccolo Funk

Cranked tight: high, dry, snapping.

Seventies Dead

Tuned low, damped with a wallet, wires loose.

Rock Rim Shot

Every hit a rim shot, pushed.

Jazz Snare

Open and ringing; play it soft for ghosts.

Loose Garage

Wires hanging off it, buzzing at pitch.

Big Eighties

A huge room, compressed, gated shut.

Init

Half head, half machine: a place to start.

MACHINES

808 Snare · 808 Snappy

Two tones and a little noise — and SNAPPY up.

909 Snare · 909 Tight

Two triangles, the drop, bright noise — tuned up and short.

707 / 727

The brittle mid-eighties Roland sample snare.

LinnDrum · DMX

A real snare, 8-bit — and bigger, roomier.

SP-1200 Crack · MPC60 Break

12-bit, pushed and filtered; a chopped break snare.

CR-78

Short, high and polite.

Simmons SDS-V

A long pitch sweep into a gated room.

808 Clap

The hand clap on its own.

TECHNO

Techno Crush

A 909 through RAZOR WING into a small room.

Warehouse Clap

Claps in a big concrete room.

Minimal Rim

A rim shot and a dotted-eighth echo.

Detroit Stack

909 snare and clap together.

Dark Room

From the far end of the room, the lights off.

House Clap Snare

A warm snare, clap on top.

PRESETS

DUB

Roots Rim Shot

A cracked high rim shot into a dotted-eighth tape echo.

Steppers

A steady snare with a quarter-note echo.

Dub Techno Space

Soft, dark; the echo does the rest.

Tubby Throw

The echo right up: a rim shot thrown into the tape.

Space Echo Clap

A clap ping-ponging on a dotted quarter.

INDUSTRIAL

Metal Shop

A ringing rim struck hard, crushed by SUPERNOVA.

Gated Machine

An electronic snare in a gated room, clipped.

EBM Snap

Square wave, tight noise, HYPERDRIVE, flattened.

Power Electronics

Loose wires, full drive, a gated wall of noise.

Factory Floor

A snare in a vast gated hall.

HARDCORE

Atari Riot

8-bit grit and SUPERNOVA: digital hardcore.

Amen Chop

The break snare: bright, 12-bit, squashed.

Breakcore Blast

RAZOR WING at the top, the compressor pumping.

Speedcore Snap

A short zap for rolls at any tempo.

Gabber Snare

A 909 snare clipped as hard as the kick.

MODERN

Trap Snare Clap

Snare and clap, stacked and tight.

DnB Snare

Bright, high, all crack.

Dubstep Snare

Snare, clap and a gated room, pushed.

Lo-Fi Ghost

Dusty and dark; play it soft.

LEVEL-MATCHED

Every preset is levelled by measurement: –15 dB RMS over the first 150 ms, and no clean snare above –1 dBFS — so flicking through compares drums, not volumes. The machine names are in the spirit of the originals; nothing is sampled from them.

SPECIFICATIONS

Format	Windows VST3, 64-bit, and a standalone application
Type	Instrument / Drum. MIDI in, stereo out
Sound	Fully synthesised. No samples
Controls	31: Head, Wires, Hit, Era, Echo, Transient, Drive, Comp, Out
Articulations	Snare, rim shot, cross-stick, clap
Keys	FIXED, GM KIT (38 / 40 / 37 / 39), CHROMATIC
Polyphony	Four overlapping hits; the oldest fades in 3 ms
Drive engines	IDLE BURN, HYPERDRIVE, RAZOR WING, SUPERNOVA
Presets	44 factory in seven banks, plus Init
User patches	Documents\Brokild patches\Snare Tactics
Sample rates	Any. Bench-tested at 44.1, 48 and 96 kHz
Licence	AGPLv3. Source at github.com/peterboggild/BrokildApps
Build	260926.4

MEASURED

Tuning against TUNE	within 0.01 cents
Oversampling	4x, always
Aliasing, RAZOR WING at 100 %	-88 dB
DECAY against its -60 dB time	within 6 %
SIZZLE against its -60 dB time	within 2 %
Echo time against the tempo	within 0.6 ms
Drive level change	within 4.1 dB
Latency at 48 kHz	95 samples, 2 ms
CPU, everything on, 8 hits/s	about 10 % of a core

Latency is reported to the host and compensated. Every number here comes from the bench in the source, which renders real audio and measures it: 72 engine checks, 72 in a plugin host, 12 on the panel.

CREDITS

NOW GO AND HIT SOMETHING

This VST comes from the curved mind of Professor Brokild, and was programmed with VScode, ChatGPT and Claude. Use of this is on your own peril. It may generate unique, beautiful, scary or useless sounds for you, and it may crash your DAW project. Have fun, play unsafely, unwisely, and unboringly.

Its sibling is **Kickstart**, the kick drum synthesiser. The demos on the website play the two together.

Cheers, Peter Bøggild, September 2026.

Drum machine names are trademarks of their owners, used only to describe the sound a preset is in the spirit of. Snare Tactics contains no samples of them.

BROKILD

Free, and free to pass on.
peterboggild.github.io/BrokildApps

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